

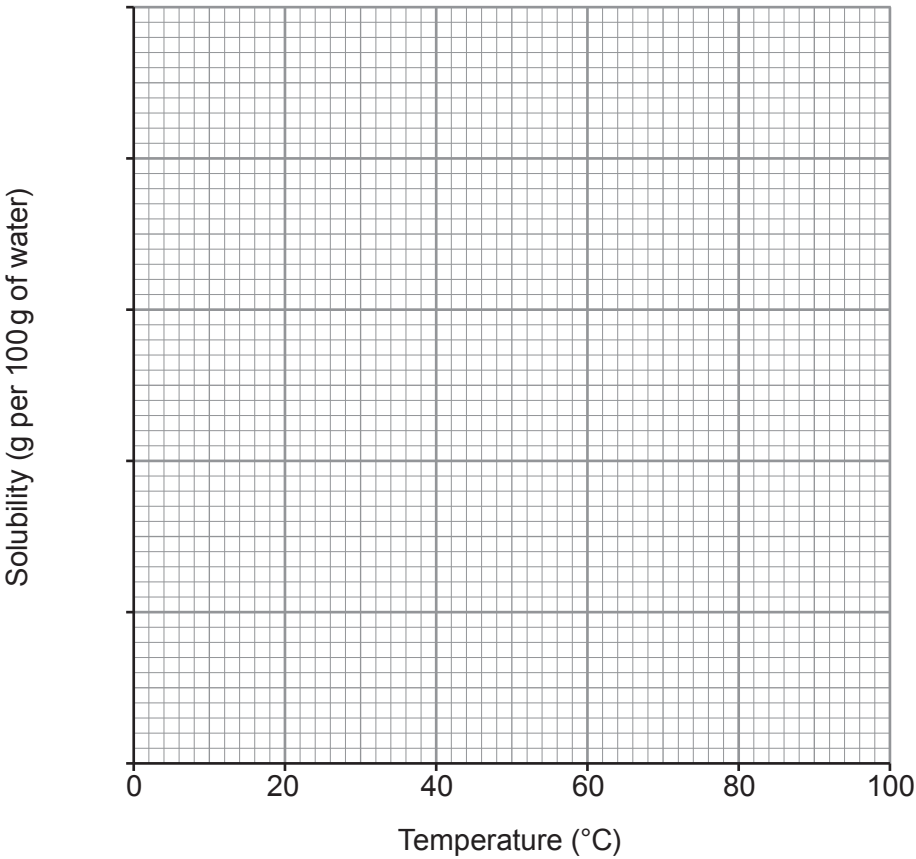
4. (a) The table shows the solubility of potassium nitrate in water at different temperatures.

Temperature (°C)	Solubility (g per 100 g of water)
0	13
20	32
40	64
60	110
80	169
100	246

(i) Choose a suitable scale for the y-axis and plot the data on the grid.

Draw a line of best fit.

[4]

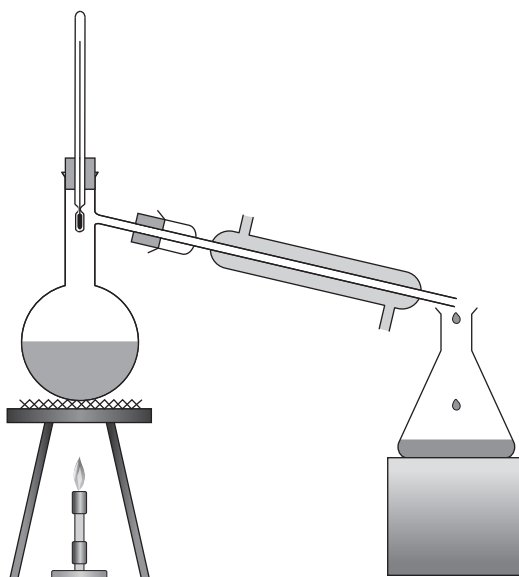


- (ii) Calculate the mass in grams of potassium nitrate that would be crystallised if 250 cm^3 of a saturated solution were cooled from 80°C to 30°C .

[3]

Mass = g

- (b) Ethanol can be separated from water by distillation using the apparatus shown.



Explain how ethanol and water are separated by distillation.

[2]

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